

# MoCHII User Guide

## Abstract

How to build and run MoCHII (MOnte Carlo for HII regions): requirements, the complete input-parameter reference, output-file formats, worked examples keyed to the regression tests, and the runbook for adding an ion. The physics and validation are documented separately in docs/MoCHII\_physics.pdf; the atomic-data fitting pipeline in docs/MoCHII\_fitting.pdf.

## Contents

|                                                  |           |
|--------------------------------------------------|-----------|
| <b>1. Requirements and build</b>                 | <b>1</b>  |
| <b>2. Input parameters</b>                       | <b>2</b>  |
| 2.1 Photons and RNG . . . . .                    | 2         |
| 2.2 Grid and dust density . . . . .              | 3         |
| 2.3 Ionizing/FUV band and the source . . . . .   | 3         |
| 2.4 Initial state and iteration . . . . .        | 4         |
| 2.5 Thermal balance and metals . . . . .         | 5         |
| 2.6 I-front re-refinement (G4) . . . . .         | 6         |
| 2.7 Dust in the band and dust emission . . . . . | 6         |
| 2.8 Peel-off imaging . . . . .                   | 7         |
| 2.9 Output . . . . .                             | 7         |
| <b>3. Output files</b>                           | <b>7</b>  |
| 3.1 <base>_rates.h5 . . . . .                    | 7         |
| 3.2 Text outputs . . . . .                       | 8         |
| 3.3 Reading outputs in Python . . . . .          | 8         |
| <b>4. Worked examples (the regression tests)</b> | <b>9</b>  |
| <b>5. Adding an ion (the G5 runbook)</b>         | <b>10</b> |
| <b>6. Practical notes</b>                        | <b>10</b> |

## 1. Requirements and build

- Intel oneAPI MPI Fortran (mpiifort/mpiifx; GNU mpif90 supported by the Makefile), HDF5 (default prefix /data/opt/hdf5\_intel; override with make HDF5\_PREFIX=...), CFITSIO.
- The SEDust dust-emission library: SEDust\_lib/ in this tree holds libsedust.a and its .mod files (copied from the MoCafe v2.00 build). Only needed at link time; the SEDust *data* directory is referenced at run time (Section 2.7).

- Python 3 with numpy/scipy (h5py, astropy, matplotlib, PyNeb for the test and pipeline scripts).

Build:

```
cd MoCHII
make          # -> MoCHII.x    (always a full clean rebuild)
make DEBUG=1  # bounds/traceback build
```

The Makefile intentionally performs a full rebuild every time (the MoCafe policy): there is no inter-module dependency tracking, and stale `.mod` files after editing `define.f90` otherwise cause runtime namelist errors.

Run:

```
mpirun -np 8 ./MoCHII.x input.in
```

MoCHII is MPI-only; the octree and all leaf arrays live in MPI-3 shared memory (one copy on each node).

## 2. Input parameters

All input is one `&parameters` namelist assigning fields of `par`. Parameters not listed keep the defaults shown. Stage switches compose: `gas_niter = 0` gives a G0 single pass (rates only); adding `gas_niter > 0` iterates the ionization (G1); `solve_te` adds the thermal balance (G2); `use_metals` the trace-metal registry; `diffuse_field` the explicit diffuse field (G3, forces case A); `refine_front` the I-front re-refinement (G4); `ion_add_dust` the dust coupling.

### 2.1 Photons and RNG

| parameter  | default | meaning                          |
|------------|---------|----------------------------------|
| no_photons | 1e5     | source packets in each iteration |
| no_print   | 1e7     | progress-print stride            |
| iseed      | 0       | RNG seed (0 = from clock/pid)    |

## 2.2 Grid and dust density

| parameter            | default              | meaning                                                                                                                                                                                                               |
|----------------------|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| grid_type            | 'car'                | 'amr' (octree) or 'uniform' (single-level leaf list stored in raster order; traversal by integer-stepping DDA, no tree or neighbor table in memory; refine_front unavailable)                                         |
| amr_file             | "                    | generic AMR leaf file (FITS/HDF5/text): columns x, y, z, level, nH [, metallicity, xHI, ndust]                                                                                                                        |
| distance_unit        | 'kpc'                | 'kpc'/'pc'/'au'/'" (code units)                                                                                                                                                                                       |
| dust_model           | 'global_dgr'         | leaf grey opacity: 'none', 'global_dgr' ( $n_{\text{H Cext}} \cdot \text{DGR}$ ), 'from_file' (ndust column), 'laursen09' (static xHI column), 'laursen09_live' (computed $x_{\text{HI}}$ , refreshed each iteration) |
| cext_dust            | 1.6059e-21           | reference extinction per H [cm <sup>2</sup> ] at lambda_ref                                                                                                                                                           |
| DGR, Z_global, Z_ref | 1e-2, 0.0134, 0.0134 | dust-model scalings                                                                                                                                                                                                   |
| f_ion_dust           | 0.01                 | laursen09 survival of dust in ionized gas                                                                                                                                                                             |
| f_pah, f_ion_pah     | 0, 0                 | PAH share of the reference extinction and its own ionized-gas survival (laursen09_live)                                                                                                                               |

## 2.3 Ionizing/FUV band and the source

| parameter                    | default     | meaning                                                                                                                                                                                                                                                                                       |
|------------------------------|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| use_ion_band                 | F           | must be T (the MoCHII transport)                                                                                                                                                                                                                                                              |
| nnu_ion                      | 16          | log-spaced bins; gates use 32; raise when widening the band                                                                                                                                                                                                                                   |
| eion_min, eion_max           | 13.598, 100 | ionizing band edges [eV]                                                                                                                                                                                                                                                                      |
| add_fuv                      | F           | FUV extension: extra log bins on [efuv_min, eion_min] below the unchanged ionizing bins; transports FUV grain heating and FUV photoionization of low-threshold metals (Mg I, C I, S I, Fe I); luminosity stays the IONIZING luminosity ( $Q_{\text{H}}$ preserved, the FUV part rides on top) |
| efuv_min                     | 6.0         | FUV lower edge [eV]                                                                                                                                                                                                                                                                           |
| nnu_fuv                      | 8           | FUV bins (16 recommended with metals)                                                                                                                                                                                                                                                         |
| tstar                        | -999        | Planck source temperature [K]                                                                                                                                                                                                                                                                 |
| ion_spectrum                 | "           | alternative 2-column spectrum file (E [eV], $L_E$ arb.)                                                                                                                                                                                                                                       |
| luminosity                   | 1.0         | BAND luminosity [erg/s]                                                                                                                                                                                                                                                                       |
| xs_point, ys_point, zs_point | 0           | point-source position (code units, recentered box)                                                                                                                                                                                                                                            |

## 2.4 Initial state and iteration

| parameter             | default | meaning                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------------|---------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| xHI_init              | 1.0     | initial $n_{\text{HI}}/n_{\text{H}}$ (an xHI file column wins); an ionized start (1e-3) converges much faster than neutral                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| xHeI_init, xHeII_init | 1, 0    | initial He fractions                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| He_abund              | 0.1     | $n_{\text{He}}/n_{\text{H}}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| gas_niter             | 0       | iterations (0 = G0 rates-only pass)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| gas_tol               | 1e-3    | convergence tolerance on $x_{\text{HII}}$ (measure set by conv_crit)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| conv_crit             | 'cell'  | 'cell': cell-max tests ( $\max  \Delta x_{\text{HII}} $ — stalls at the front-cell Monte Carlo noise floor $\sim 2\text{--}4\text{e-}2$ at $4\text{e}7$ packets); 'vol': volume-integrated tests $\sum V  \Delta x_{\text{HII}}  / \sum V x_{\text{HII}} < \text{gas\_tol}$ and $\sum V n_e  \Delta T_e  / \sum V n_e T_e < \text{gas\_tol\_te}$ (recommended; noise-robust, and the $n_e$ weight removes vacuum and no-heating cells). The vol measures floor at their own Monte Carlo level ( $\sim 3\text{e-}3$ and $7\text{e-}3$ at $8\text{e}6$ packets, scaling as $1/\sqrt{N}$ ) — set the tolerances just above it |
| ion_relax             | 1.0     | under-relaxation on $x$ ( $< 1$ damps front oscillation)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| case_ab               | 'B'     | case A/B recombination ('A' forced by the diffuse field)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

## 2.5 Thermal balance and metals

| parameter        | default       | meaning                                                                                                                                                                                                                                                                                                           |
|------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| solve_te         | F             | bisection on $T_e$ coupled with the ionization                                                                                                                                                                                                                                                                    |
| te_fixed         | 1e4           | fixed $T_e$ [K] when solve_te is off                                                                                                                                                                                                                                                                              |
| te_min, te_max   | 1e3, 5e4      | bisection bracket [K]                                                                                                                                                                                                                                                                                             |
| gas_tol_te       | 1e-3          | convergence on $\max  \Delta T_e /T_e$                                                                                                                                                                                                                                                                            |
| atomic_dir       | 'data/atomic' | coefficient files (relative to the run directory)                                                                                                                                                                                                                                                                 |
| use_metals       | F             | registry cascade + Tier-1 metal cooling                                                                                                                                                                                                                                                                           |
| abund_C/N/O/Ne/S | HII20 values  | $n(X)/n(H)$ ; an element is active when $> 0$                                                                                                                                                                                                                                                                     |
| abund_Ar/Mg/Fe   | 0             | further registry elements (gas-phase values; Mg/Fe are heavily depleted)                                                                                                                                                                                                                                          |
| abund_Si/Cl/Ca   | 0             | five-stage registry elements (gas-phase; Si and especially Ca are heavily depleted, Cl nearly undepleted — Orion-like values $3e-6 / 3e-7 / 2e-8$ ); adds [Si II] 34.8 $\mu m$ , [Si III] 1883/1892, [Si IV] 1394/1403, [Cl II], the [Cl III] 5518/5538 density pair, [Cl IV], [Ca II], [Ca V]                    |
| ion_metal_abs    | T             | metal photoionization absorption in the band opacity (active with use_metals); the only gas opacity in the FUV bins                                                                                                                                                                                               |
| emis_output      | F             | write _emis (HDF5/FITS): leaf-by-leaf line emissivities + the state needed for maps (Section 3.)                                                                                                                                                                                                                  |
| diffuse_field    | F             | explicit ground-recombination packets (G3)                                                                                                                                                                                                                                                                        |
| hei_diffuse      | F             | fourth diffuse channel (needs diffuse_field): He I excited-level recombination photons (19.82 eV, 584 Å, two-photon) — correct energy bookkeeping; moves Te by $\sim 1\%$ at HII40, He split unchanged                                                                                                            |
| gaunt_vh14       | F             | free-free Gaunt factors from the van Hoof et al. (2014) table (data/gauntff_vh14.dat) instead of Hummer (1988); the two agree to $\lesssim 0.5\%$ in $T_e$                                                                                                                                                        |
| metal_ne         | F             | PDR: electrons from the metal cascade join the $n_e$ closure (the only electrons beyond the I-front)                                                                                                                                                                                                              |
| metal_heat       | F             | PDR: metal photoheating in the thermal balance                                                                                                                                                                                                                                                                    |
| grain_pe         | F             | PDR thermal package: grain photoelectric heating + grain recombination cooling (Bakes & Tielens 1994) from the local FUV field (needs add_fuv), plus H-impact [C II] 158 $\mu m$ / [O I] 63 $\mu m$ cooling; grain terms scaled by $pe\_scale \times$ the leaf dust amount relative to MW (from the leaf opacity) |
| pe_scale         | 1.0           | photoelectric-efficiency scale factor                                                                                                                                                                                                                                                                             |

## 2.6 I-front re-refinement (G4)

| parameter                 | default | meaning                                             |
|---------------------------|---------|-----------------------------------------------------|
| refine_front              | F       | rebuild the octree on the I-front once              |
| refine_iter               | 20      | at which iteration                                  |
| refine_lbase, refine_lmax | 5, 7    | base/front levels                                   |
| refine_eps                | 1e-2    | front flag: $\epsilon < x_{\text{HI}} < 1-\epsilon$ |
| refine_dx                 | 0.2     | or a face-neighbor $x_{\text{HI}}$ jump above this  |

## 2.7 Dust in the band and dust emission

| parameter                | default         | meaning                                                                                                                                                                                                                                                                                                                                        |
|--------------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ion_add_dust             | F               | dust absorption in the band (dusty Strömgren)                                                                                                                                                                                                                                                                                                  |
| ion_dust_kext            | "               | kext table with EUV coverage (e.g. data/kext_albedo_WD_MW_3.1_60_D03.all_2009); required                                                                                                                                                                                                                                                       |
| ion_dust_scatter         | F               | sample dust scattering (HG, albedo weights)                                                                                                                                                                                                                                                                                                    |
| dust_sed                 | F               | SEDust stochastic dust SED at output time                                                                                                                                                                                                                                                                                                      |
| dust_model_sed           | 'astrodust'     | 'astrodust'/'dl07'/'zubko'                                                                                                                                                                                                                                                                                                                     |
| sed_workdir              | "               | a SEDust sed/ tree (dielectric tables are read relative to it), e.g. the MoCafe copy .../MoCafe_v2.00/SEDust/sed                                                                                                                                                                                                                               |
| sed_qtable, sed_sizedist | MoCafe defaults | optics/size-distribution tables (relative to sed_workdir)                                                                                                                                                                                                                                                                                      |
| sed_NT, sed_Tlo, sed_Thi | 200, 2.7, 5e3   | SEDust temperature grid                                                                                                                                                                                                                                                                                                                        |
| dust_emis_bands          | unset           | up to 8 wavelengths [ $\mu\text{m}$ ] (e.g. 8, 24, 70, 160): SEDust band emissivities of each leaf $\rightarrow$ _dustemis (blocks emis_dust/wl_dust [ $\text{erg s}^{-1} \text{cm}^{-3} \mu\text{m}^{-1}$ ]); the Python map maker projects them into dust-band images (the IR is optically thin)                                             |
| sed_pah_live             | F               | $x_{\text{HII}}$ -dependent PAH survival in the emission: the leaf spectrum is assembled from the model channels with the PAH channel weighted by $(x_{\text{HI}} + f_{\text{ion,pah}} x_{\text{HII}})$ (needs a model with a 'PAH' channel, i.e. astrodust); turns the 8- $\mu\text{m}$ map into the classic PAH ring at the ionization front |

## 2.8 Peel-off imaging

| parameter        | default  | meaning                                                                                                                                                                                                                                                                                                                   |
|------------------|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ion_peel         | F        | after convergence, one extra transport pass peels direct (stellar + diffuse) and dust-scattered contributions toward the observers $\rightarrow$ <code>_image</code> (blocks <code>direct_ion/scatt_ion</code> , plus <code>_fuv</code> with <code>add_fuv</code> ) [ $\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$ ] |
| peel_bins        | F        | bin-resolved cubes <code>direct_cube/scatt_cube</code> ( $n_x, n_y, n_\nu$ ) in addition to the band-integrated channel images (hardness maps, synthetic EUV/FUV photometry)                                                                                                                                              |
| obsx, obsy, obsz | unset    | observer positions (or directions) in code units; or use <code>alpha/beta/gamma</code> angle lists [deg]; default: one observer on +z far away                                                                                                                                                                            |
| distance         | auto     | observer distance (code units)                                                                                                                                                                                                                                                                                            |
| nxim, nyim       | 129, 129 | image pixels                                                                                                                                                                                                                                                                                                              |
| dxim, dyim       | auto     | pixel scale [deg]; auto-sized to cover the box                                                                                                                                                                                                                                                                            |

## 2.9 Output

`out_file` (base name; extension follows `file_format = 'hdf5' or 'fits'`).

## 3. Output files

### 3.1 <base>\_rates.h5

One dataset for each extension, leaf-ordered:

| extension                     | content                                                                                |
|-------------------------------|----------------------------------------------------------------------------------------|
| Gamma_HI/HeI/HeII             | photoionization rates [ $\text{s}^{-1}$ ]                                              |
| Heat_HI/HeI/HeII              | photoheating per particle [erg/s]                                                      |
| Heat_dust                     | grain heating [ $\text{erg s}^{-1} \text{cm}^{-3}$ ] (when <code>ion_add_dust</code> ) |
| T_dust                        | equilibrium mixture dust temperature [K]                                               |
| x_HI, x_HeI, x_HeII, n_e, T_e | converged state                                                                        |
| x_<el>_stages                 | metal ion fractions (stage, leaf)                                                      |
| J_nu                          | band mean intensity (bin, leaf) [ $\text{erg/s/cm}^2/\text{Hz/sr}$ ]                   |
| E_bin, dE_bin                 | band grid [eV]                                                                         |
| LeafXYZ                       | leaf centers (code units)                                                              |

### 3.2 Text outputs

| file         | content                                                                                                                                                                                                                                                                                                                                   |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| _lines.txt   | line luminosities: SH95 H I and He II recombination lines (1640, 4686, ... when an He III zone exists) + Tier-2 collisional lines; columns elem, stage, $\lambda[\text{\AA}]$ , $L$ , $L/L(\text{H}\beta)$                                                                                                                                |
| _emis.h5     | (with emis_output) leaf-by-leaf line emissivities: emis_h (SH95 lines) and emis_<el>_<stage> blocks [ $\text{erg s}^{-1} \text{cm}^{-3}$ ; $\sum \text{emis} \cdot V = L$ , rows match _lines.txt], wl_* wavelengths [ $\text{\AA}$ ], plus LeafXYZ, LeafSize, n_H, n_e, T_e, x_HI/HeI/HeII, x_<el>_stages — self-contained for line maps |
| _nebcont.txt | nebular continuum $L_\nu$ (fb+ff tables, Milne, two-photon)                                                                                                                                                                                                                                                                               |
| _dustir.txt  | equilibrium-mixture dust IR spectrum (1–1000 $\mu\text{m}$ ) with the energy-closure header                                                                                                                                                                                                                                               |
| _dustsed.txt | SEDust stochastic dust SED with PAH bands (when dust_sed)                                                                                                                                                                                                                                                                                 |
| _dustemis.h5 | (with dust_emis_bands) SEDust band emissivities of each leaf, same block layout as _emis — readable by EmisData for dust-band maps                                                                                                                                                                                                        |
| _image.h5    | (with ion_peel) peel-off images for each observer: direct_ion/scatt_ion (+_fuv) [ $\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$ ], suffix _obs<k> when several observers                                                                                                                                                              |

### 3.3 Reading outputs in Python

tools/python/mochii\_output.py is an importable module (works directly in a notebook) that reads every section file (HDF5 or FITS; read\_sections), the line table (read\_lines\_table), and the text spectra (read\_spectrum), and builds 2D maps from the emissivity file:

```
import sys; sys.path.insert(0, ".../MoCHII/tools/python")
import matplotlib.pyplot as plt
from mochii_output import EmisData

d = EmisData("run_emis.h5")
d.info()                # blocks, fields, line count
d.line_list()           # every stored line (block, wl)

# ready-made figures (notebook): returns (fig, ax)
fig, ax = d.plot_line("o_3", 5007.)          # log S map
fig, ax = d.plot_line("h", 6563., unit="intensity",
                    photons=True) # photons/s/cm^2/sr
fig, ax = d.plot_field("T_e", weight="EM")
fig, axs = plt.subplots(1, 2); d.plot_line("s_2", 6717., ax=axs[0]) \
    ; d.plot_line("s_2", 6731., ax=axs[1]) # doublet side by side
```



```
# raw arrays, when the figure is yours to build
img, ext = d.line_map("o_3", 5007., axis="z", npix=512)
img, ext = d.line_map("h", 6563., unit="intensity", photons=True)
img, ext = d.field_map("T_e", weight="EM") # LOS-weighted average
Q = d.line_luminosity("h", 4861., photons=True) # photon rate [1/s]
```

Line maps are flux-conserving: each leaf's luminosity is deposited with exact area overlap onto the pixel grid (AMR leaf sizes handled), so  $\sum S A_{\text{pix}} = L_{\text{line}}$  to machine precision. `unit='flux'` (default) gives surface brightness [ $\text{erg s}^{-1} \text{cm}^{-2}$ ]; `unit='intensity'` gives specific intensity [ $\text{erg s}^{-1} \text{cm}^{-2} \text{sr}^{-1}$ ] =  $S/4\pi$  (isotropic emission, optically thin); `photons=True` converts either to photon units (divides by  $hc/\lambda$  of the line). A `slab=(lo,hi)` cut restricts the line of sight. Field maps average `T_e`, `n_e`, `x_HI`, ... along the line of sight with weight 'EM' ( $n_e^2 V$ ), 'ne', 'nH', or 'V'. The same module doubles as a command-line tool for quick looks:

```
python3 mochii_output.py run_emis.h5 # list lines
python3 mochii_output.py run_emis.h5 --line o_3 5007 \
    --npix 512 --log --out o3.png
python3 mochii_output.py run_emis.h5 --line h 6563 \
    --unit intensity --photons # photons/s/cm^2/sr
python3 mochii_output.py run_emis.h5 --field T_e --weight EM
```

## 4. Worked examples (the regression tests)

Each `tests/` directory is self-contained: grid builder or a symlinked grid, the input file, and a `check_*.py` gate script.

| directory                 | what it demonstrates / gate                                 |
|---------------------------|-------------------------------------------------------------|
| <code>g0_gamma</code>     | rate integrals vs. analytic attenuation (bias +0.004%)      |
| <code>g1_stromgren</code> | Strömgren sphere, case B fixed $T_e$ ; AMR $\equiv$ uniform |
| <code>g2a_thermal</code>  | thermal H/He nebula vs. the 1D exact reference              |
| <code>g2_hii</code>       | MOCASSIN Lexington HII20/HII40 benchmarks + line fluxes     |
| <code>g4_refine</code>    | I-front re-refinement vs. native level 7                    |
| <code>g4_tng</code>       | Illustris-TNG post-processing demo (shadows, maps)          |
| <code>d_dusty</code>      | dusty Strömgren; scattering bracket; $T_d/\text{IR}$        |

A minimal thermal + metals + dust input (from `d_dusty`):

&parameters

```
par%no_photons = 4.0e7      par%iseed = 1234
par%grid_type = 'amr'      par%amr_file = 'amr_L6.fits'
par%distance_unit = 'pc'   par%dust_model = 'global_dgr'
par%use_ion_band = .true.  par%nnu_ion = 32
par%eion_min = 13.598      par%eion_max = 100.0
```

```

par%tstar = 4.0e4                par%luminosity = 3.178e38
par%xHI_init = 1.0e-3           par%He_abund = 0.1
par%gas_niter = 60              par%gas_tol = 2.0e-3
par%solve_te = .true.          par%atomic_dir = '../data/atomic'
par%use_metals = .true.
par%ion_add_dust = .true.
par%ion_dust_kext = '../data/kext_albedo_WD_MW_3.1_60_D03.all_2009'
par%cext_dust = 4.868e-22       par%DGR = 1.0
par%out_file = 'run.h5'         par%file_format = 'hdf5'
/

```

## 5. Adding an ion (the G5 runbook)

Adding an ion touches only data:

1. `tools/fitting/fit_cooling_tier1.py`: add the ion with a  $T_i$  ladder guessed from its `.elvlc` level energies; run; check the reported max error and the overlay figure.
2. `tools/fitting/fit_nlevel_tier2.py`: add the ion with the level count needed for its diagnostics; run.
3. `tools/fitting/make_element_data.py`: add the element to `ELEMENTS` ( $Z$ , stage count) and its thresholds; charge exchange entries where sources exist; run. The CHIANTI `.rrparams/.drparams` types 1–3 are handled (RR/RR2/DR/DR2 lines).
4. Fortran: a `par%abund_<El>` field and one entry in the `species_setup` name/abundance lists (only for a NEW element; new stages of a known element need nothing).
5. Verify: extend `verify_nlevel_pyneb.py` with a diagnostic ratio where PyNeb has data; run a smoke test and check the new lines in `_lines.txt`.

Data caveats met so far: Ar II has no CHIANTI v11 level/collision data (stage tracked without lines); Fe III collision strengths differ from PyNeb's BB14 by up to ~16% in 4658/4702 (known spread for iron); charge-exchange stage labels differ between the Huang (2023) and MOCASSIN transcriptions for Mg/Fe (the Huang labels are adopted; see `make_element_data.py`).

## 6. Practical notes

- **Bins**: 32 bins give ~2% rate-integral discretization; use 64 for sub-percent work. The `add_fuv` extension carries its own bins below `eion_min`, so the ionizing resolution is unaffected.
- **FUV and the PDR**: without `add_fuv` there is no radiation beyond the I-front and the metals there recombine toward neutral; with it, FUV penetrates the neutral gas (dust extinction only) and produces the PDR-side C II

and Mg II. Metal photoheating and grain photoelectric heating are not in the thermal balance, so PDR-zone temperatures are indicative only.

- **Convergence:** use `conv_crit = 'vol'` for production — the cell-max criterion stalls at the front-cell noise floor (and, with `add_fuv`, at no-heating cells where  $T_e$  pins to `te_min`). Both measures are printed each iteration. The G3 diffuse field smooths the front and converges far better.
- **Starts:** prefer an ionized initial state; a neutral start advances the front  $\sim$ one cell per iteration.
- **text tables:** must cover the band (the astrodust table stops at 912 Å; use the D03 table for EUV work). Tables are sorted on read (the D03 file has an out-of-order seam).
- **Re-refinement:** one event per run; the old shared-memory windows are reclaimed only at exit.
- **SEDust:** `sed_workdir` must point at a SEDust `sed/` tree; the exact stochastic solve costs  $\sim 0.05$ – $0.1$  s for each gas leaf (distributed over ranks).